

- 30 The rest masses of three particles, tritium ${}^3_1\text{H}$, proton ${}^1_1\text{p}$, and neutron ${}^1_0\text{n}$, are 3.0160 u, 1.0086 u and 1.0097 u respectively. A tritium may disintegrate to its constituent protons and neutrons if it
- A captures a photon of energy 10 MeV.
 - B captures a photon of energy 12 MeV.
 - C emits a photon of energy of 10 MeV.
 - D emits a photon of energy of 12 MeV.

Ans: B

$$\begin{aligned}\text{Energy required} &= (3.016 - 1.0086 - (2 \times 1.0097))u \\ &= 11.2 \text{ MeV}\end{aligned}$$

End of Paper